



EMOTICONS

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INTRODUCTION

- Multiple embedding strategies (Word2Vec, GloVe, FastText, BERT)
- Multiple topic modeling techniques (LDA, BERTopic)
- 2 datasets (emotag, Tweets with emojis)

Goals:

- Evaluate the coherence between annotated emotion scores and textual definitions.
- Visualize and interpret emotional proximities across different embeddings.
- Compare performance of classical and transformer-based models in capturing emotive semantics.
- Investigate antagonistic relationships (e.g., joy–sadness) in lexical and distributional contexts.
- Predict the use of emojis



1F60D

Joy 0.83



1F494

Sadness 0.94



1F440

Anticipation 0.81



1F389

Surprise 0.50



1F624

Disgust 0.64



1F628

Fear 0.97



1F621

Anger 1.00



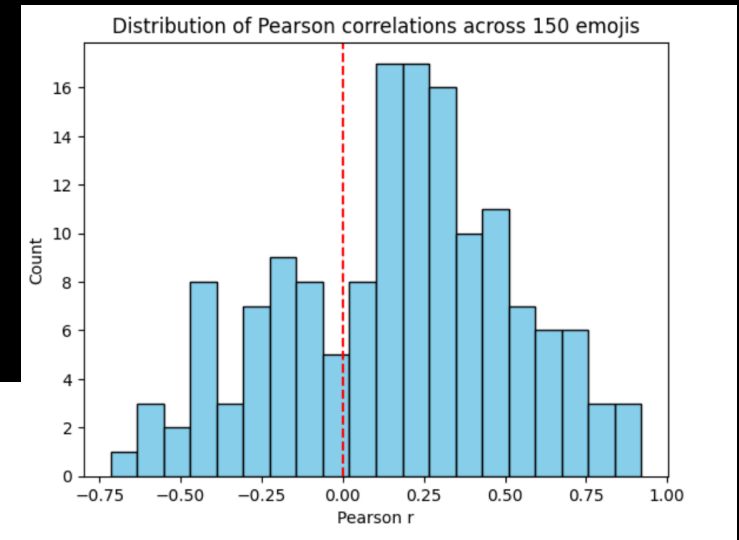
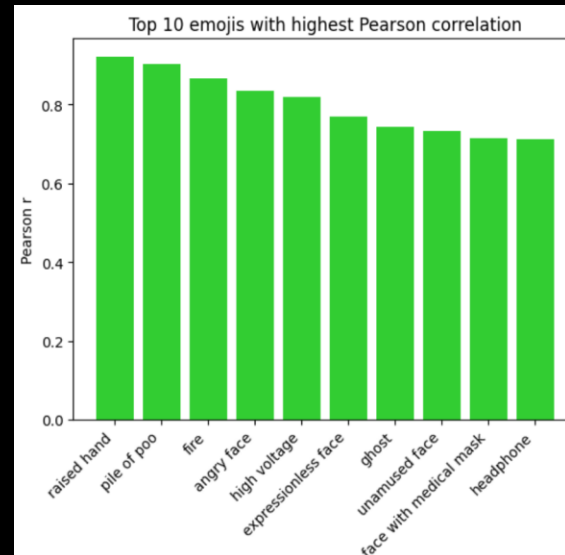
1F339

Trust 0.72

WORD2VEC-BASED WORD EMBEDDING

Goal: Compare emoticon's definitions and EmoTag1200 labeling.

- Emoticon definition from [cheatsnake/emojihub · GitHub](https://github.com/cheatsnake/emojihub).
- Convert definition to embedding vectors using word2vec.
- Compute similarity between each emotions and definition vector using cosine similarity
- Combine 8 emotion similarities as an 8 dimension vector.
- Compare with EmoTag using Pearson Correlation.



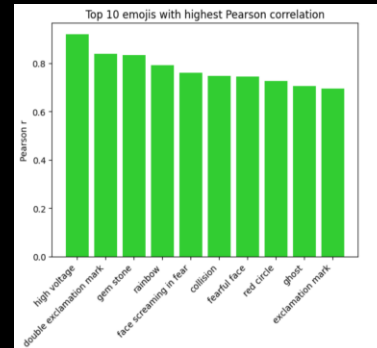
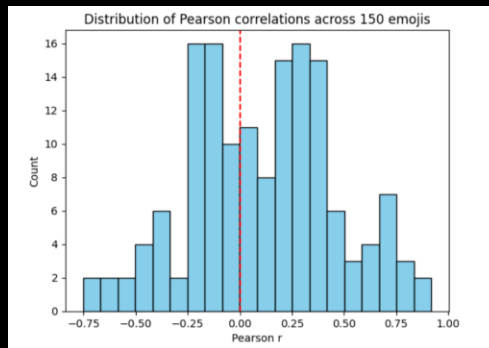
	emoji	pearson_r	p_value
0	rainbow	0.271190	0.515899
1	crescent moon	-0.296659	0.475538
2	new moon face	0.202063	0.631317
3	sun with face	0.353910	0.389747
4	glowing star	0.371975	0.364211

Mean r = 0.1674455991222437
Median r = 0.2050820759336442

OTHER EMBEDDING MODELS

Goal: Compare similarity results of the previous task across different embedding models

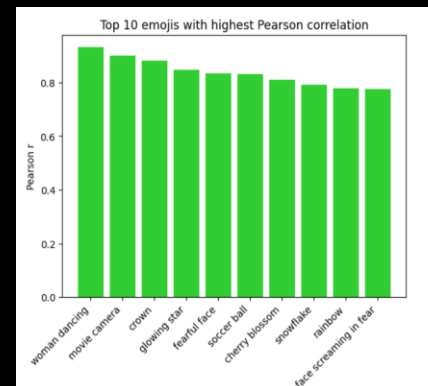
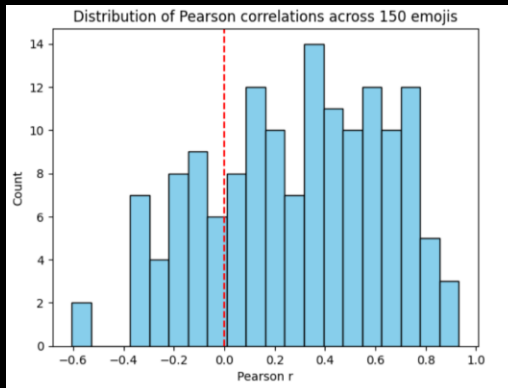
- GloVe model:



	emoji	pearson_r	p_value
0	rainbow	0.792307	0.019054
1	crescent moon	0.271716	0.515053
2	new moon face	-0.421242	0.298632
3	sun with face	-0.020211	0.962116
4	glowing star	0.447241	0.266536

Mean r = 0.11050485239900176
Median r = 0.10315479462983432

- FastText model:



	emoji	pearson_r	p_value
0	rainbow	0.777787	0.023063
1	crescent moon	0.310311	0.454439
2	new moon face	-0.162926	0.699876
3	sun with face	-0.048512	0.909183
4	glowing star	0.848535	0.007730

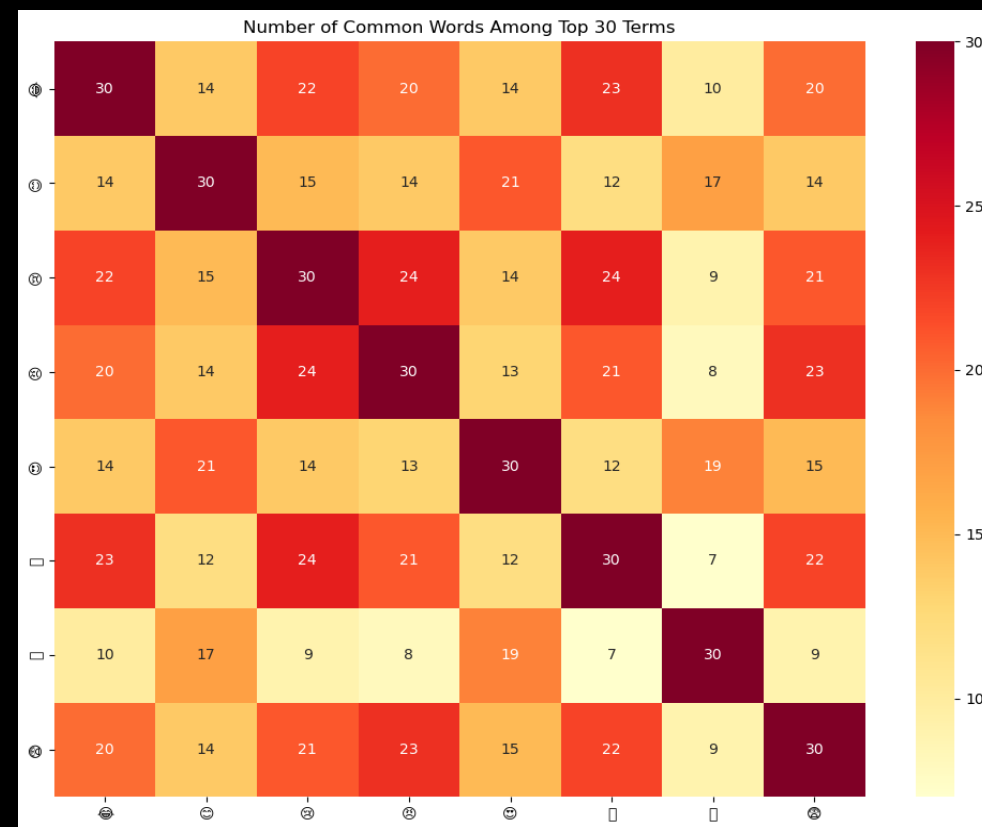
Mean r = 0.2870921779982472
Median r = 0.3321520522757172

KAGGLE TWEETS WITH EMOJIS

Goal: Find 30 most frequent words associated with each emojis and create a common associated words matrix

- Corpus per emoji (~20k tweets)
- Preprocessing takes out stopwords

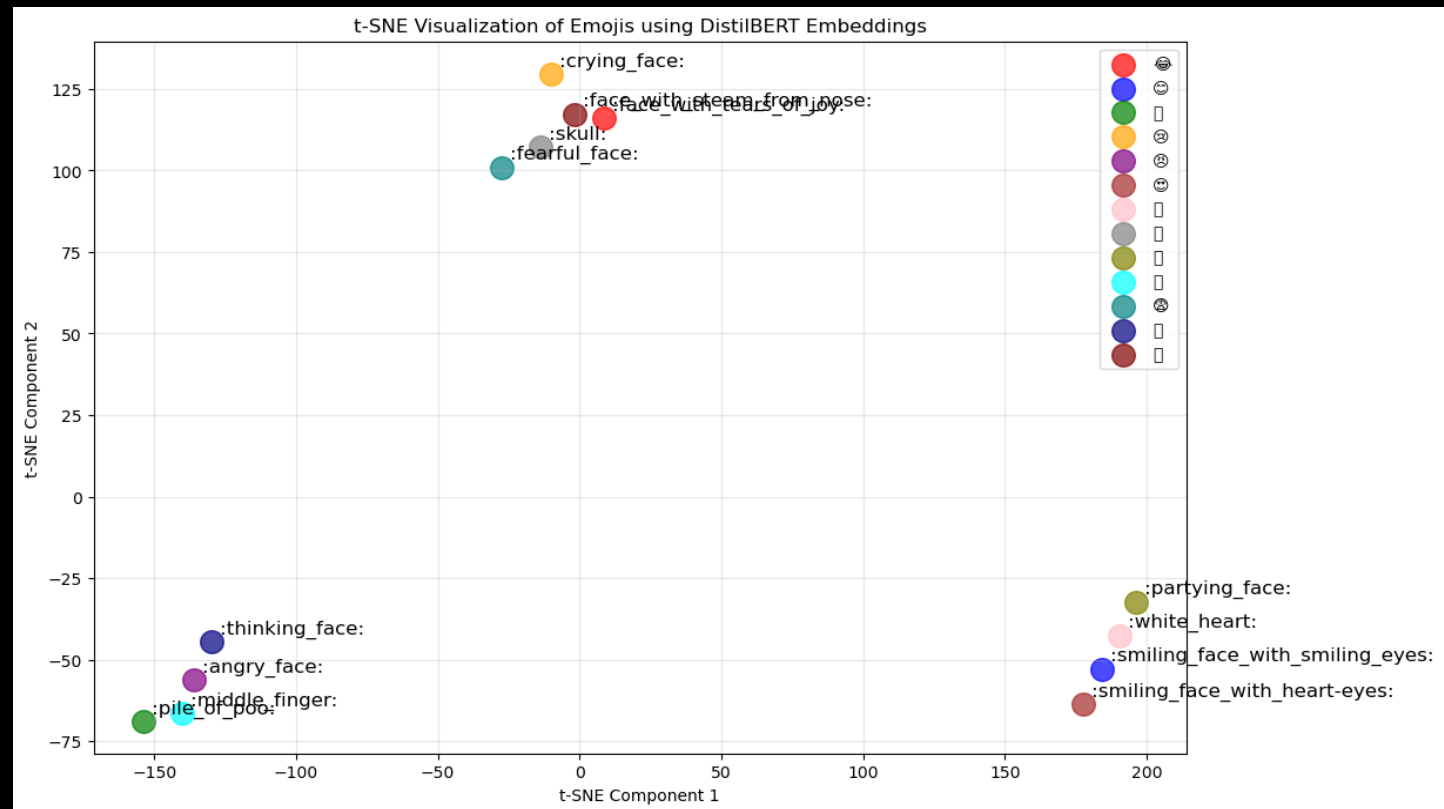
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😄 top 5 terms: [('like', 1602), ('dont', 1083), ('one', 983), ('get', 913), ('know', 828)]
😊 top 5 terms: [('thank', 2136), ('good', 1515), ('like', 1288), ('day', 1150), ('love', 1127)]
😏 top 5 terms: [('like', 2362), ('dont', 1183), ('one', 928), ('get', 896), ('cant', 851)]
😬 top 5 terms: [('like', 1503), ('dont', 1398), ('people', 1283), ('get', 1146), ('amp', 1107)]
😂 top 5 terms: [('love', 2287), ('beautiful', 1312), ('easter', 1145), ('happy', 1087), ('like', 930)]
💀 top 5 terms: [('like', 2444), ('dont', 1251), ('one', 1107), ('even', 1081), ('get', 925)]
🎉 top 5 terms: [('happy', 4893), ('birthday', 4158), ('day', 2138), ('amp', 1509), ('easter', 1257)]
😱 top 5 terms: [('like', 1871), ('dont', 1054), ('one', 864), ('know', 705), ('get', 703)]
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DISTILBERT EMBEDDING

Goal: Represent how close are the emojis using TSNE when using the DistilBERT embedding

- 40% smaller than BERT
- Reduction of dimensions via TSNE



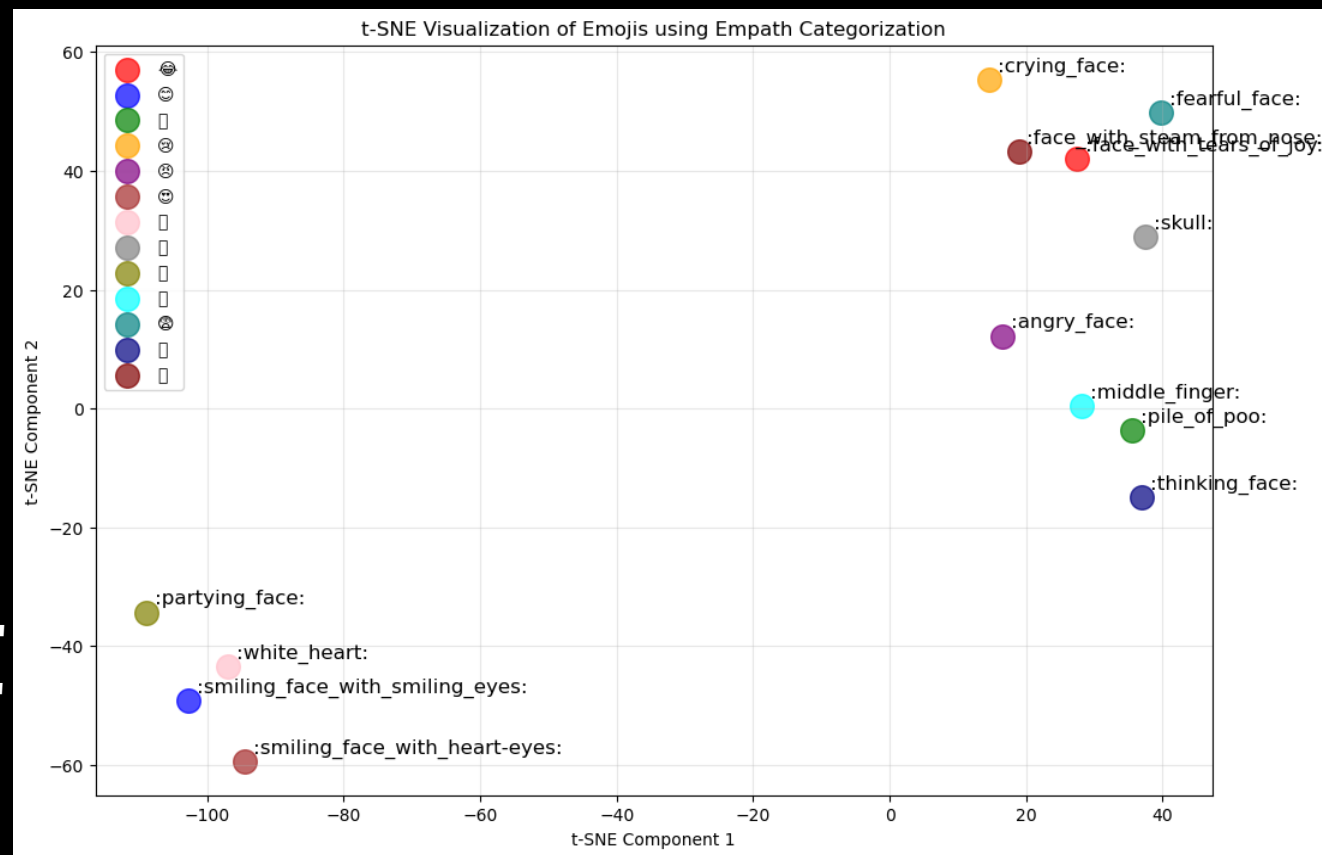
EMPATH CATEGORIZATION EMBEDDING

Goal: Represent how close are the emojis using TSNE when using the DistilBERT embedding

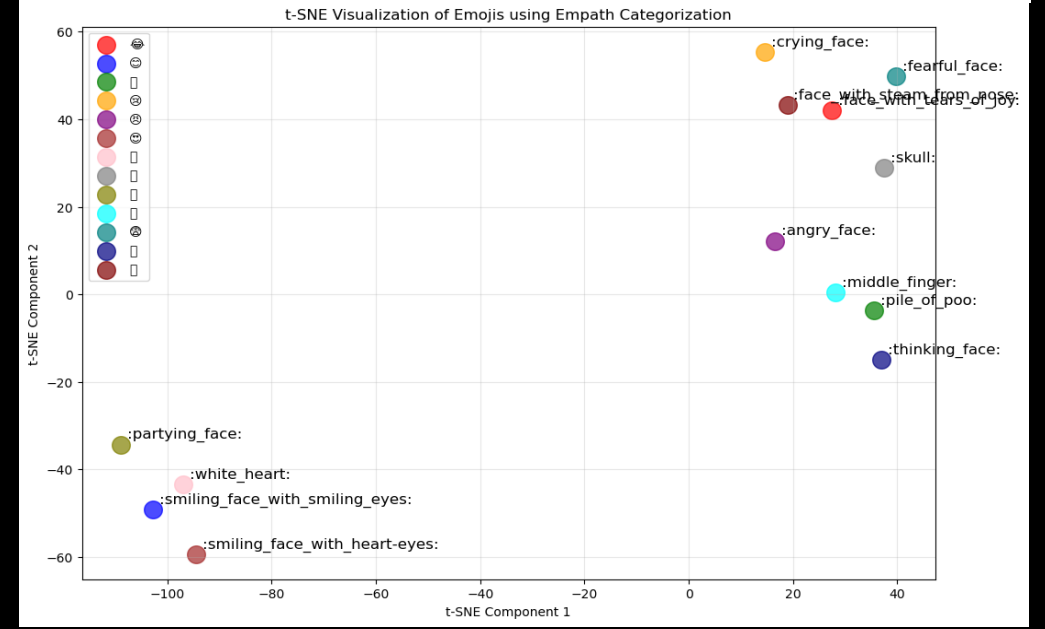
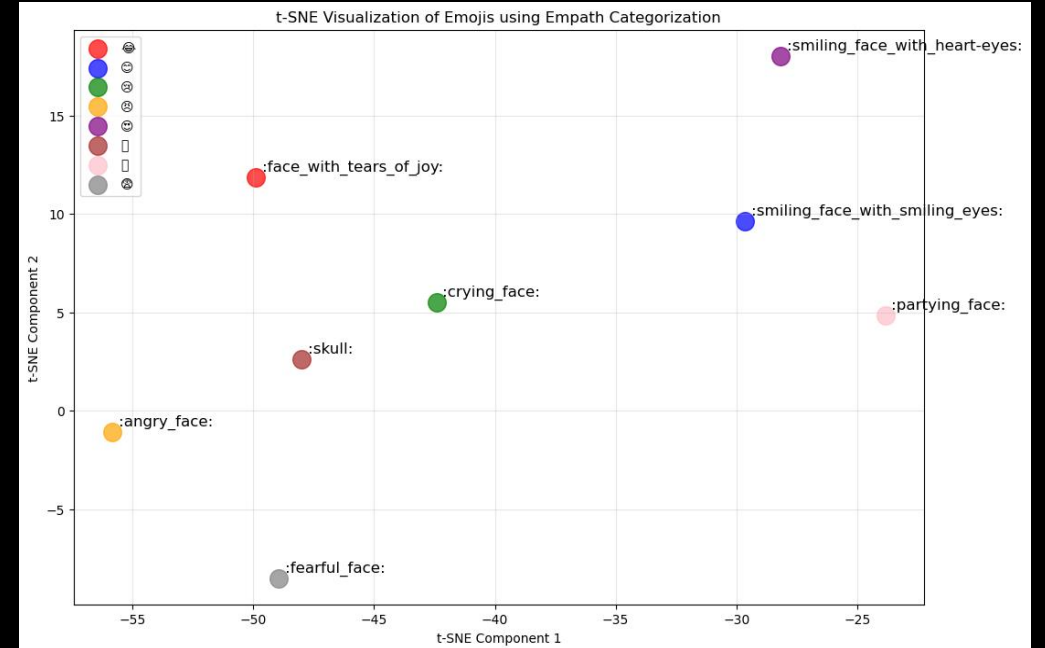
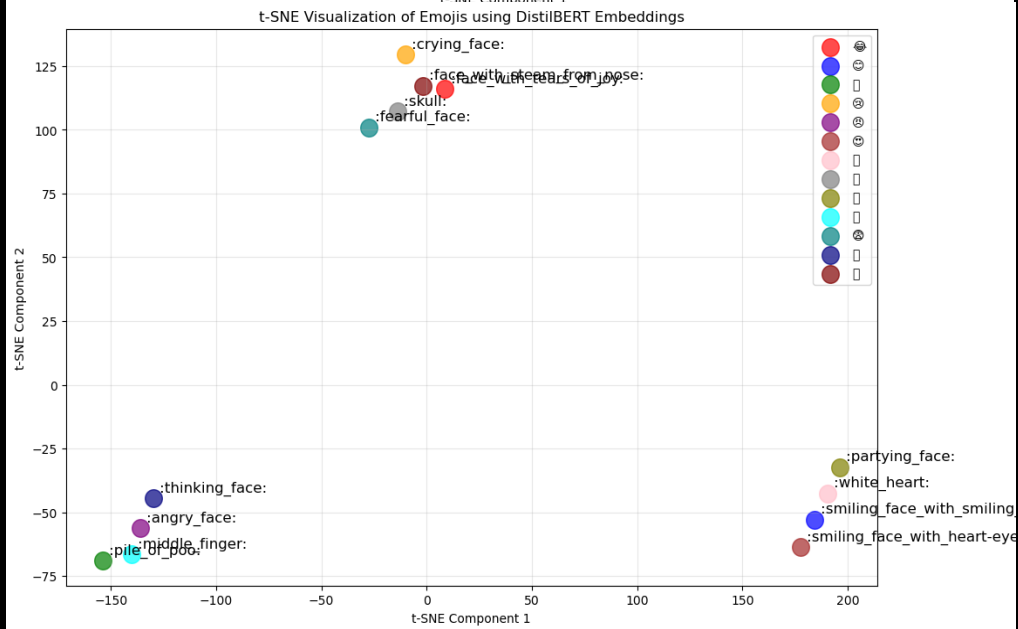
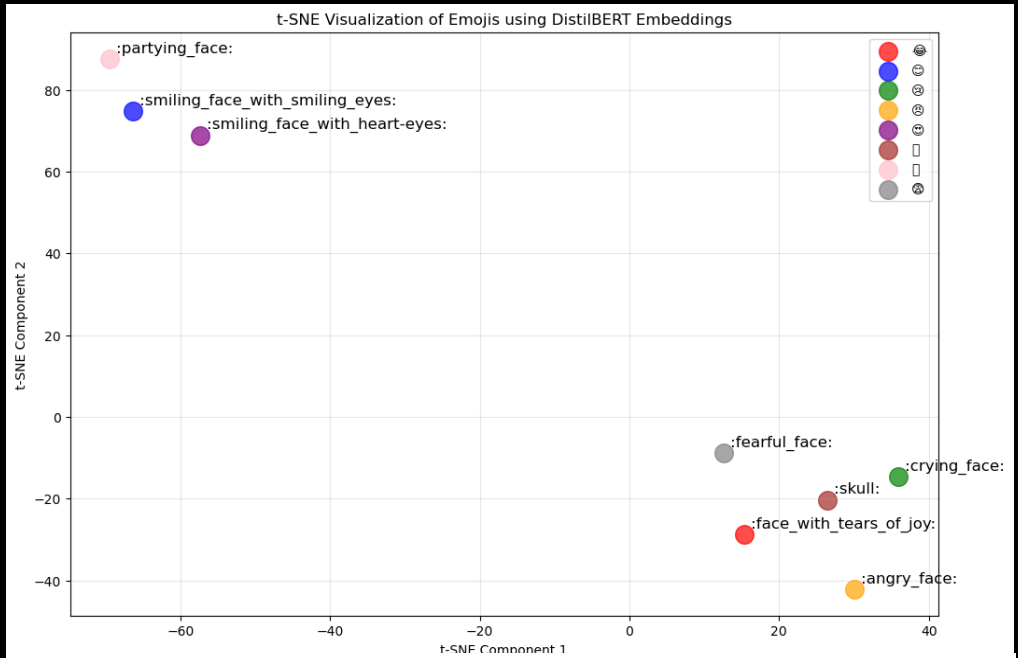
- TSNE reduction of dimensions
- Empath has a list of 194 categories
- Empath gives a score to each

Category for each emoji

Empath categories: ['achievement', 'affection', 'aggression', 'air_travel', 'alcohol', 'ancient', 'anger', 'animal', ...]



COMPARISON OF DISTILBERT AND EMPATH



LDA-BASED TOPIC MODELLING

Goal: infer emotional content of emoji-specific tweet corpora via unsupervised topic modelling

- Corpus per emoji (~20k tweets)
- Preprocessing: remove URLs, mentions, punctuation, stopwords
- Apply **LDA (1 topic)** → extract **8 top words**
- Map each word to emotions using cosine similarity (GloVe embeddings)
- Compare with *EmoTag1200* emotion distributions

Emoji	Top 8 LDA keywords	Cosine	Pearson
🔥 fire	one, like, love, get, easter, time, see, happy	0.90	0.37
😓 grinning_face_with_sweat	like, one, get, know, time, think, people, even	0.87	0.43
😨 fearful_face	like, one, people, get, know, see, would, time	0.83	0.38
👍 thumbs_up	good, great, like, well, one, get, see, thanks	0.80	0.24
✨ sparkles	happy, easter, love, like, thank, day, time, one	0.80	0.62
💀 skull	like, one, even, bro, get, people, know, got	0.79	0.27
👻 ghost	like, one, love, good, ghost, get, day, time	0.76	0.64
🎉 party_popper	happy, birthday, congratulations, day, easter, amp, good, party	0.76	0.72

BERTOPIC-BASED TOPIC MODELLING

Motivation: LDA ignores context → use transformer-based embeddings

- Embeddings from SentenceTransformer ("all-MiniLM-L6-v2")
- Topics learned with BERTopic (contextual clustering)
- Same mapping to emotion space
- Cosine similarity with EmoTag profiles

Emoji	Top 8 BERTopic keywords	Cosine
🔥 fire	love, like, time, got, good, today, let, new	0.90
😓 grinning_face_with_sweat	like, time, know, good, people, think, day, really	0.88
😨 fearful_face	like, know, people, going, day, said, got, god	0.82
👍 thumbs_up	good, like, great, thank, thanks, know, time, see	0.81
💀 skull	like, know, people, shit, think, got, time, life	0.80
🙏 folded_hands	thank, good, god, know, love, day, bless, like	0.77
🎉 party_popper	birthday, happy, congratulations, day, thank, love, great, best	0.75
😊 smiling_face	like, love, happy, time, thank, day, great, good	0.71
❤️ red_heart	love, day, happy, good, thank, easter, time, one	0.68
💩 pile_of_poo	like, know, people, bull, piece, good, think, bad	0.55

DEEPMOJI-BASED EMOJI PREDICTION FROM TEXT

Goal: inverse task → predict emoji from text

- Use HuggingFace TorchMoji pretrained model
- Input: tweet text → Output: most probable emojis
- Evaluate % of tweets correctly mapped to their label emoji

Emoji	Top@1	Top@2	Top@3	Top@4	Top@5
👍 thumbs_up	0.35	0.50	0.60	0.66	0.71
😉 winking_face	0.31	0.48	0.57	0.63	0.68
🙏 folded_hands	0.29	0.35	0.39	0.43	0.46
😋 face_savoring_food	0.19	0.23	0.25	0.27	0.28
😍 smiling_face_with_heart_eyes	0.14	0.20	0.25	0.29	0.32
😂 face_with_tears_of_joy	0.09	0.15	0.21	0.27	0.33
👁 eyes	0.09	0.14	0.16	0.19	0.22
💀 skull	0.06	0.10	0.13	0.16	0.19
😎 smiling_face_with_sunglasses	0.05	0.09	0.14	0.19	0.26
✨ sparkles	0.05	0.07	0.10	0.12	0.14
😭 loudly_crying_face	0.02	0.06	0.12	0.16	0.20
❤ red_heart	0.02	0.06	0.13	0.19	0.26
😓 grinning_face_with_sweat	0.02	0.04	0.06	0.09	0.12
😊 smiling_face	0.01	0.03	0.06	0.09	0.12
🤧 face_with_steam_from_nose	0.00	0.02	0.07	0.11	0.14

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03/02/2026

```
$ python3 text_emojize.py --text="I am pleased to present this research report on emoticons!"
```

→ I am pleased to present this research report on emoticons! 😊👍😄😁😜

```
$ python3 text_emojize.py --text="It really annoys me when Bert takes three hours to find his topics!"
```

→ It really annoys me when Bert takes three hours to find his topics!

😡😡😡😡👊

ANTONYM-BASED EVALUATION OF EMOJI OPPOSITION

Goal: test *semantic opposition* between emotional emojis (😭 vs 😊)

- Extract 20 most frequent words per emoji corpus (after extended stopwords)
- Filter words with ≥ 2 antonyms (WordNet)
- Compute overlap of antonym pairs

Sadness 😭: ['first', 'bad', 'real', 'getting', 'show', 'take', 'made', 'whole', 'girl', 'new', 'miss', 'come', 'give', 'better', 'let', 'hard', 'best', 'went', 'end', 'little']

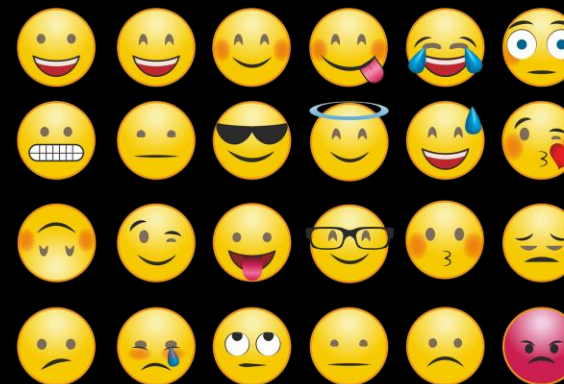
Happiness 😊: ['new', 'best', 'work', 'first', 'welcome', 'take', 'better', 'little', 'made', 'friend', 'come', 'lets', 'friends', 'keep', 'getting', 'sure', 'give', 'let', 'long', 'start']

$\text{AntOverlap}(\text{😭} \rightarrow \text{😊}) = 0.15$

$\text{AntOverlap}(\text{😊} \rightarrow \text{😭}) = 0.40$

take $\leftrightarrow$ give, went $\leftrightarrow$ come,
best $\leftrightarrow$ bad, first $\leftrightarrow$ end, keep $\leftrightarrow$ let

CONCLUSION



Fairly good match between emojis and the emotion conveyed in a text,
which can even be predicted

BUT we encountered certain limitations:

- Emojis found in the dataset are not identical
- Pre-processing is made more difficult due to the specific vocabulary used in tweets
- Emojis in tweets are sometimes ironic, which can mislead the interpretation of emotion (irrelevant topics)